Walker Anderson

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Skills

- **Power Electronics:** DC/DC converters (buck, flyback), pulsed power, magnetics design, control loop and transient analysis, switching devices, component stress/thermal analysis, high-voltage validation
- **Simulation:** LTspice, SaberRD, SIMetrix/SIMPLIS, Simulink, MATLAB, Monte Carlo, Worst-case analysis
- **Software:** Python, C++, KiCAD, Altium, Xpedition Designer, Microsoft suite
- **Hardware:** High Voltage testing, Oscilloscopes, Programmable power supplies, Electronic loads

Technical Projects

Pulsed Electromagnetic Coil

- Designed a 500V, 4.23μF capacitor-discharged pulsed magnetic coil system targeting 100 mT field generation, using a 48A peak current produced by an SCR switched resonant pulse.
- Modeled coil inductance, stored energy, and transient LC discharge behavior using first-principles analysis and circuit simulation to predict peak current and pulse dynamics.
- Designed measurement architecture using Hall-effect sensing and galvanically isolated signal chains to capture high di/dt transients, leveraging integrated current sensing ICs with low conductor loss and high bandwidth.
- Simulated system power converter stages (flyback capacitor charger, auxiliary supplies) and pulsed LC discharge system in LTspice to validate operating conditions, component stress, and transient response prior to hardware implementation.

Wireless Power Transfer

- Designed and built a series-resonant inductive wireless power transfer system delivering 100W across an 8-inch air gap at 60% efficiency, converting 51V, 3.1A input to 22V, 4.4A output.
- Developed planar coil geometry (10 turns, 12-inch diameter) with 62 μH measured inductance and tuned series resonance at 100 kHz to maximize coupling over large air gaps.
- Designed a half-bridge MOSFET driver with bootstrap gate drive, controlled by an MCU-generated switching signal to excite the primary resonant coil.
- Characterized system performance using oscilloscope and DMM measurements, evaluating voltage/current waveforms, efficiency, and sensitivity to coil alignment and spacing.

Experience

January 2025 - Present **Raytheon RMD** - Electrical Engineer I

- Designed and analyzed DC/DC power converters (buck, flyback) for EMI-sensitive applications, including control loop stability, transient response, and regulation performance.
- Performed simulation-driven validation using LTspice (Monte Carlo, worst-case, corner analysis) and conducted component stress analysis to ensure robust operation across tolerances and operating conditions.
- Developed analog sensing and diagnostics circuits (voltage monitoring, level shifting, signal conditioning) and performed error budget analysis to quantify measurement accuracy.
- Executed hardware validation and bring-up of high-voltage systems, including safe-to-turn-on procedures, lab debugging with oscilloscopes/power equipment, and thermal characterization of switching devices in phase-shifted full-bridge converters.

Education

May 2024 **University of Arizona**

Bachelor of Science: Electrical and Computer Engineering - Cum Laude

May 2018 **Arizona State University**

Bachelor of Science: Industrial Design